

JINWOO KIM

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Research Interest

My focus is in making deep learning models generalize better beyond their training data so that they can be used to solve challenging problems, such as those in scientific domains. I am studying this problem from the complementary perspectives of invariance and equivariance, parallel neural processing, and Solomonoff induction. I often use tools from theories of graphs, manifolds, (semi)groups and categories, as well as Markov processes such as random walks, diffusions, and flows. Recently, I am focused on flow language models, latent reasoning, synthetic data, and vision pretraining.

Academic Placement and Employment

KAIST, Postdoctoral Researcher	03/2026 – 02/2027
New York University, Visiting Scholar (host: Kyunghyun Cho, Rajesh Ranganath)	11/2025 – 01/2026
LG AI Research, Research Intern (host: Moontae Lee, Honglak Lee)	01/2022 – 07/2022

Education

KAIST, M.S./Ph.D. in Computer Science (advisor: Seunghoon Hong) - Ph.D. Thesis: Architecture-Agnostic Invariances for Deep Learning KAIST CoE Best Ph.D. Dissertation Award	03/2021 – 02/2026
KAIST, B.S. in Brain Engineering and Computer Science (double major) - GPA 4.05/4.3 (Summa Cum Laude)	03/2016 – 02/2021

Publications

C: conference, J: journal, W: workshop, P: preprint, *: equal contribution, †: equal advising

P6. Look Back: Reinforcement Learning with Counterfactual World Models

Jinwoo Kim^{*}, Toghrul Abbasov^{*}, Lucius E.J. Bynum, Aahlad Manas Puli, Stephanie Milani, Hyunwoo Ryu, Seunghoon Hong, Kyunghyun Cho[†], Rajesh Ranganath[†]
Under review

P5. Why Does Rewriting Pretraining Data Improve Generalization?

Jiyun Park^{*}, Jinwoo Kim^{*}, Kyungbin Min, Hoyeon Chang, Sohee Yang, Kyunghyun Cho[†], Seunghoon Hong[†]
Under review

P4. Large Flow Language Models

Arnav Chavan^{*}, Chanhyuk Lee^{*}, Udbhav Bamba^{*}, Jaehoon Yoo, Seunghoon Hong, Aditi Raghunathan[†], Jinwoo Kim[†]
Under review

P3. Asynchronous Flow Maps for Flow Language Models

Chanhyuk Lee, Wonjung Kim, Patrick Pynadath, Jaehoon Yoo, Satoshi Hayakawa, Ruqi Zhang, Jinwoo Kim[†], Seunghoon Hong[†]
Under review

P2. Generalization in Continuous Flows for Categorical Data

Manan Agarwal, Jinwoo Kim
Under review

P1. Thinking with Looped Flows

Ayhan Suleymanzade, Chanhyuk Lee, Floor Eijkelboom, Nicholas M. Boffi, İsmail İlkan Ceylan[†], Jinwoo Kim[†]
arXiv 2026

C17. MUX: Continuous Reasoning via Multiplexed Tokens

Ayhan Suleymanzade, Halil Alperen Gozeten, Michael Bronstein, İsmail İlkan Ceylan[†], Jinwoo Kim[†]
NeurIPS 2026 **Spotlight (292/30709=0.95%)**

- C16. Flow Map Language Models: One-step Language Modeling via Continuous Denoising**
 Chanhyuk Lee, Jaehoon Yoo, Manan Agarwal, Sheel Shah, Jerry Huang, Aditi Raghunathan, Seunghoon Hong, Nicholas M. Boffi[†], [Jinwoo Kim](#)[†]
 NeurIPS 2026 / COLM 2026 NonAR-LM Workshop **Spotlight**
- C15. Multi-View Relational Distillation for Spatial Reasoning with Vision-Language Models**
 Kiet T. Nguyen, Hanbo Shim, [Jinwoo Kim](#)[†], Seunghoon Hong[†]
 NeurIPS 2026
- C14. RelAgent: LLM Agents as Data Scientists for Relational Learning**
 Xingyue Huang, Louis Tichelman, [Jinwoo Kim](#), Krzysztof Olejniczak, İsmail İlkan Ceylan
 NeurIPS 2026
- W6. To Learn is to Wander: Learning Across Graphs and Tasks with Random Walks**
 Louis Tichelman, Xingyue Huang, [Jinwoo Kim](#), İsmail İlkan Ceylan
 NeurIPS 2026 GDDL Workshop
- W5. One-step Generation via Riemannian Wasserstein Gradient Flows**
 David Li, Chanhyuk Lee, Jaehoon Yoo, Nikita Gushchin, Floor Eijkelboom, Eric Moulines, Maxim Panov, Alexander Korotin, Seunghoon Hong[†], [Jinwoo Kim](#)[†]
 NeurIPS 2026 DiffuLM Workshop
- W4. Self-conditioned Flow Map Language Models via Fixed-point Flows**
 Jaehoon Yoo*, Wonjung Kim*, Floor Eijkelboom, Chanhyuk Lee, Nicholas M. Boffi, Seunghoon Hong[†], [Jinwoo Kim](#)[†]
 COLM 2026 NonAR-LM Workshop **Oral**
- W3. Posterior Refinement: Fast Language Generation via Any-Order Flow Maps**
 Manan Agarwal*, Sheel Shah*, Chanhyuk Lee, Jaehoon Yoo, Jerry Huang, Seunghoon Hong, Aditi Raghunathan, [Jinwoo Kim](#)[†], Nicholas M. Boffi[†]
 COLM 2026 NonAR-LM Workshop **Spotlight**
- W2. What are the Right Symmetries for Formal Theorem Proving?**
 Krzysztof Olejniczak, Radoslav Dimitrov, Xingyue Huang, Bernardo Cuenca Grau, [Jinwoo Kim](#)[†], İsmail İlkan Ceylan[†]
 ICML 2026 AI4Math Workshop
- C13. Inverting Data Transformations via Diffusion Sampling**
[Jinwoo Kim](#)*, Sékou-Oumar Kaba*, Jiyun Park, Seunghoon Hong[†], Siamak Ravanbakhsh[†]
 ICML 2026
- C12. Flock: A Knowledge Graph Foundation Model via Learning on Random Walks**
[Jinwoo Kim](#)*, Xingyue Huang*, Krzysztof Olejniczak, Kyunghbin Min, Michael Bronstein, Seunghoon Hong, İsmail İlkan Ceylan
 ICLR 2026
- C11. Sequence Modeling with Spectral Mean Flows**
[Jinwoo Kim](#), Max Beier, Petar Bevanda, Nayun Kim, Seunghoon Hong
 NeurIPS 2025
- C10. Revisiting Random Walks for Learning on Graphs**
[Jinwoo Kim](#), Olga Zaghen*, Ayhan Suleymanzade*, Youngmin Ryou, Seunghoon Hong
 ICLR 2025 **Spotlight (380/11672=3.26%)** / **ELLIS Mobility Grant, ICML 2024 GRaM Workshop**
- C9. 3D Denoisers are Good 2D Teachers: Molecular Pretraining via Denoising and Cross-Modal Distillation**
 Sungjun Cho, Dae-Woong Jeong, Sung Moon Ko, [Jinwoo Kim](#), Sehui Han, Seunghoon Hong, Honglak Lee, Moontae Lee
 AAAI 2025 **Oral**
- C8. Simulation-Free Training of Neural ODEs on Paired Data**
 Semin Kim*, Jaehoon Yoo*, [Jinwoo Kim](#), Yeonwoo Cha, Saehoon Kim, Seunghoon Hong
 NeurIPS 2024
- W1. Learning Symmetrization for Equivariance with Orbit Distance Minimization**
 Tien Dat Nguyen*, [Jinwoo Kim](#)*, Hongseok Yang, Seunghoon Hong
 NeurIPS 2023 NeurReps Workshop

C7. **Learning Probabilistic Symmetrization for Architecture Agnostic Equivariance**
Jinwoo Kim, Tien Dat Nguyen, Ayhan Suleymanzade, Hyeokjun An, Seunghoon Hong
NeurIPS 2023 **Spotlight (378/12345=3.06%)**

C6. **Universal Few-shot Learning of Dense Prediction Tasks with Visual Token Matching**
Donggyun Kim, Jinwoo Kim, Seongwoong Cho, Chong Luo, Seunghoon Hong
ICLR 2023 **Outstanding Paper Award (4/4955=0.08%)** / **Silver Prize, Samsung Humantech Paper Award, 2023**

C5. **Pure Transformers are Powerful Graph Learners**
Jinwoo Kim, Tien Dat Nguyen, Seonwoo Min, Sungjun Cho, Moontae Lee, Honglak Lee[†], Seunghoon Hong[†]
NeurIPS 2022

C4. **Transformers meet Stochastic Block Models: Attention with Data-Adaptive Sparsity and Cost**
Sungjun Cho, Seonwoo Min, Jinwoo Kim, Moontae Lee, Honglak Lee, Seunghoon Hong
NeurIPS 2022

C3. **Equivariant Hypergraph Neural Networks**
Jinwoo Kim, Saeyoon Oh, Sungjun Cho, Seunghoon Hong
ECCV 2022

C2. **Transformers Generalize DeepSets and Can be Extended to Graphs and Hypergraphs**
Jinwoo Kim, Saeyoon Oh, Seunghoon Hong
NeurIPS 2021 / **Qualcomm Innovation Fellowship Korea, 2021**

C1. **SetVAE: Learning Hierarchical Composition for Generative Modeling of Set-Structured Data**
Jinwoo Kim*, Jaehoon Yoo*, Juho Lee, Seunghoon Hong
CVPR 2021

J1. **Spontaneous Retinal Waves Can Generate Long-Range Horizontal Connectivity in Visual Cortex**
Jinwoo Kim*, Min Song*, Jaeson Jang, Se-Bum Paik
The Journal of Neuroscience 40(34) 2020

Honors

Awards

Best Ph.D. Dissertation Award, KAIST College of Engineering	2026
Outstanding Researcher Award, KAIST-Mila Prefrontal AI Research Center	2024
ELLIS Mobility Grant [C10], ICML 2024 GRaM Workshop	2024
Outstanding Paper Award [C6], ICLR 2023	2023
Silver Prize, Samsung Humantech Paper Award [C6]	2023
Qualcomm Innovation Fellowship Korea [C2]	2022
KAIST Engineering Innovator Award (five recipients), KAIST College of Engineering	2020

Scholarships and Fellowships

Kwanjeong Education Foundation Scholarship	2022 – 2023
Korea National Science & Technology Scholarship	2018 – 2019
KAIST Alumni Fellowship	2017 – 2020
KAIST Presidential Fellowship	2016 – 2020
Hansung Scholarship for Gifted Students	2015 – 2016

Invited Talks

Flow Language Models [C11, C16, W3, W4, W5, P1, P2, P3, P4]	
• MIT CSAIL (host: Hyunwoo Ryu)	12/2026 (planned)
• POSTECH IME (host: Sangwoo Mo)	10/2026 (planned)
• Unannounced startup	09/2026
• SNU IPAI Seminar (host: Jaesik Park, Pierre Musacchio)	07/2026
• Discrete Diffusion Reading Group (host: Subham Sahoo, Zhihan Yang)	04/2026
• Ben-Gurion University of the Negev (host: Ilan Naiman)	12/2025

Architecture-Agnostic Invariances for Deep Learning [C7, W1, C10, C12, C13]	
• Mila - Quebec AI Institute (host: Minsu Kim, Junyeob Baek)	07/2025
• KAIST AI899 Geometric DL (host: Sungsoo Ahn)	05/2025
• Mila - Quebec AI Institute (host: Siamak Ravanbakhsh, Sékou-Oumar Kaba)	12/2024
• KAIST-Mila Prefrontal AI Research Center (host: Sungjin Ahn)	11/2024
• Sungkyunkwan University (SKKU) (host: Chang Woo Myung)	08/2024
• Pohang University of Science and Technology (POSTECH) (host: Sungsoo Ahn)	11/2023
Visual Token Matching [C6]	
• KAIST–Samsung Electronics DS Division Exchange Meetup (host: Chulmoo Kang)	08/2023
Transformers for Sets, Graphs, and Hypergraphs [C1, C2, C5]	
• Microsoft USA (host: Nabiha Asghar)	01/2023
• Qualcomm Korea (host: Jaewon Choi)	01/2023
• NeurIPS 2022 at KAIST (host: Dongkwan Kim)	11/2022
• Learning on Graphs and Geometry Reading Group (LoGaG) (host: Hannes Stärk)	08/2022
• KAIST AI Workshop 21/22 (host: Dongkwan Kim)	01/2022
• NeurIPS 2021 Social: ML in Korea (host: Jung-Woo Ha)	12/2021
• Naver AI Author Meetup for CVPR 2021 (host: Jung-Woo Ha)	09/2021
• Korean Conference on Computer Vision 2021 (host: Jongwoo Lim)	09/2021
Retinal Waves and Prenatal Wiring of Primary Visual Cortex [J1]	
• Society for Neuroscience, Chicago, IL, US	10/2019

Academic Services

Area Chair: ICLR 2027, ICML 2026 GFM

Conference Reviewer: NeurIPS, ICLR, ICML, CVPR, ICCV, AISTATS, LoG, TAG-DS, IJCNN, ACCV

Journal Reviewer: TMLR, Neural Networks, IJCV

Teaching

Teaching Assistant, KAIST School of Computing

Undergraduate Research Program (URP) 2022, 2024

Introduction to Deep Learning (CS492I) 2021, 2022, 2023

Computer Vision (CS576) 2022, 2023

School of Computing Colloquium (CS966, CS986) 2021

Teaching Assistant, Samsung Electronics

Samsung Research AI Expert Program 2021 – 2024

Mentoring

Chanwoo Kim, M.S. student @ KAIST 2026

David Li, Ph.D. student @ MBZUAI [W5] 2026

Toghrul Abbasov, B.S. student @ KAIST [P6] 2026

Kiet T. Nguyen, Ph.D. student @ KAIST [C14] 2026

Manan Agarwal, M.S. student @ CMU [C16, W3] 2025 – 2026

Sheel Shah, M.S. student @ CMU [C16, W3, P2] 2025 – 2026

Chanhyuk Lee, M.S. student @ KAIST [C16, W3, W4, W5, P1, P3, P4] 2025 – 2026

Kyungbin Min, B.S. student @ KAIST [C12] → M.S. student @ KAIST [P5] 2025 – 2026

Jiyun Park, B.S. student @ KAIST [C13] → M.S. student @ KAIST [P5] 2024 – 2026

Nayun Kim, B.S. student @ KAIST [C11] → M.S. student @ KAIST 2024 – 2026

Ayhan Suleymanzade, B.S. student @ KAIST [C7, C10] → Ph.D. student @ EPFL [C17, P1] 2023 – 2026

Youngmin Ryou, B.S. student @ KAIST [C10] → M.S. Student @ KAIST 2023 – 2026

Nicole Shen, B.S. student @ MIT → Intern @ MIT LIDS 2024

Semin Kim, M.S. student @ KAIST [C8] → Ph.D. student @ KAIST 2023 – 2024

Olga Zaghen, M.S. student @ UniTrento [C10] → Ph.D. student @ UvA Amsterdam 2023

Tien Dat Nguyen, B.S. student @ KAIST [C5, C7, W1] → M.S. student @ UWaterloo 2021 – 2023

Daniel Sungho Jung, B.S. student @ Penn State → Ph.D. student @ SNU	2021
Saeyoon Oh, B.S. student @ KAIST [C2, C3] → Engineer @ FuriosaAI	2021

Projects

Korea National Research Foundation (NRF), Physical Reasoning in Language Models	2024 – 2025
Korea Ministry of Science and ICT, Visual Commonsense Reasoning	2021 – 2023
Korea National Research Foundation (NRF), Cooperative Robotic Intelligence	2021 – 2023

References

Seunghoon Hong, Associate Professor at KAIST	seunghoon.hong@kaist.ac.kr
İsmail İlkan Ceylan, Associate Professor at TU Wien	ismail.ceylan@tuwien.ac.at
Nicholas M. Boffi, Assistant Professor at Carnegie Mellon University	nboffi@andrew.cmu.edu
Kyunghyun Cho, Full Professor at New York University	kyunghyun.cho@nyu.edu
Siamak Ravanbakhsh, Associate Professor at McGill University	siamak.ravanbakhsh@mcgill.ca
Moontae Lee, Head of Superintelligence Lab at LG AI Research	moontae.lee@lgresearch.ai